MODULE 7.3

Cloud Infrastructure Security

Infrastructure Layers | Shared Responsibility | Key Considerations

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Topics:

- 7.3.1 Infrastructure Security
- 7.3.2 Cloud Security Responsibilities
- 7.3.3 Other Cloud Infrastructure Security Considerations
- 7.3.4 Check Your Understanding
- Summary
- References

Infrastructure Security

- **Three layers, one provider:** cloud infrastructure is built from compute, storage, and network layers — a single provider delivers all three, but each layer needs its own distinct security controls.
- **Compute security:** hardening VM images and container images before deployment, keeping the guest OS patched, and relying on the hypervisor to isolate one tenant's workload from another's.
- **Network security:** virtual firewalls and micro-segmentation control traffic between workloads, securing the virtual network overlay that connects VMs to each other and out to the internet — the same job a physical firewall does, just software-defined.
- **Storage security:** encrypting data at rest and tightly controlling access permissions on storage buckets and volumes. Misconfigured, publicly-accessible storage is one of the most common real-world causes of cloud data breaches.

Cloud Security Responsibilities

The shared responsibility model: who manages each layer of the stack, by service model.

Layer	IaaS	PaaS	SaaS
Data	You	You	You
Applications	You	You	Provider
Runtime	You	Provider	Provider
Middleware	You	Provider	Provider
Operating System	You	Provider	Provider
Virtualization	Provider	Provider	Provider
Servers (Compute)	Provider	Provider	Provider
Storage & Networking	Provider	Provider	Provider
Physical Data Center	Provider	Provider	Provider

Teal = customer ("You") responsibility. Navy = provider responsibility. Notice the "staircase": the more complete the service (IaaS → PaaS → SaaS), the less the customer manages.

Other Cloud Infrastructure Security Considerations

- **Misconfiguration:** the single leading cause of real-world cloud breaches — even strong technical controls fail the moment a storage bucket or security group is left open by mistake.
- **Patch management:** in IaaS, the customer still owns patching the guest OS and installed applications; falling behind here remains one of the most common paths to compromise.
- **Logging and monitoring:** cloud platforms generate enormous volumes of log data — it has to be centralized and actively reviewed, or a misconfiguration or intrusion can sit unnoticed for months.
- **DDoS protection:** providers typically offer built-in mitigation at the network edge, but application-layer DDoS protection is usually still the customer's responsibility to configure.
- **Vendor lock-in:** heavy reliance on a provider's proprietary APIs and services can make it costly or technically difficult to migrate away later.
- **Data residency and sovereignty:** legal or regulatory requirements may mandate that certain data physically remain within a specific country's or region's data centers.

Check Your Understanding

Who is responsible — the customer or the provider?

Q: Patching the guest operating system on an IaaS virtual machine.
A: You (the customer)

Q: Maintaining physical security of the data center housing the servers.
A: The provider, in every service model

Q: Managing user access to the data stored inside a SaaS application.
A: You (the customer) — even in SaaS, data remains the customer's responsibility

Q: Securing the hypervisor that isolates one tenant's VM from another's.
A: The provider

Q: Configuring the application-layer logic and middleware of a PaaS-hosted app.
A: You (the customer)

Q: Leaving a storage bucket publicly accessible by mistake — whose failure is this?
A: The customer's — misconfiguration is consistently the leading cause of real-world cloud breaches

Summary

- **7.3.1 —** cloud infrastructure security covers three layers — compute, storage, and network — each needing its own controls even though one provider delivers all three.
- **7.3.2 —** the shared responsibility model follows a staircase: the more complete the service (IaaS → PaaS → SaaS), the more the provider manages and the less the customer does — but data ownership never fully leaves the customer.
- **7.3.3 —** beyond the stack itself, misconfiguration, patch management, and centralized logging/monitoring are the operational practices that determine whether a cloud deployment is actually secure day to day.
- **The throughline:** moving to the cloud shifts who manages a layer, but it never eliminates the need to manage it — a gap in understanding exactly where the provider's responsibility ends is exactly where breaches happen.

References

Cisco Networking Academy — Network Defense, Module 7.3: Cloud Infrastructure Security

https://www.netacad.com

Cloud Security Alliance — Shared Responsibility Model Guidance

https://cloudsecurityalliance.org/research

NIST — The NIST Definition of Cloud Computing (SP 800-145)

https://csrc.nist.gov/publications/detail/sp/800-145/final